A Unified Network for Detecting Out-Of-Context Information Using Generative Synthetic Data

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ABSTRACT

In our modern world, the manipulation of digital content, especially the usage of out-of-context images known as Cheapfakes, has become a significant challenge for the endorsement of integrity and trustworthiness for information on the Internet. This highlights an urgent need for effective detection of the misuse of images and their accompanied captions. Motivated by this issue, our research presents an innovative approach to the solution of this problem. Participating in the ACM ICMR 2024 Grand Challenge on Detecting Cheapfakes, we leverage a unified end-to-end network, integrated with generative synthetic data for training. After complete evaluation, our proposed network demonstrated a remarkable accuracy of 95.60% on the public test dataset for Task 1, as well as efficiency in Task 2. This paper highlights the notable potential of employing an end-to-end network for Cheapfakes detection, which composes a significant contribution to the advancement of multimedia content integrity. Our source code is publicly available at https://github.com/thanhson28/cheapfakes_detection_icmr2024.git

CCS CONCEPTS

• **Security and privacy → Security services**; **Social aspects of security and privacy**; • **Computing methodologies → Scene understanding**; **Neural networks**; *Natural language processing*; *Image representations.*

KEYWORDS

Cheapfakes Detection, Visual Entailment, Out-of-context, Misinformation, Miscontextualization

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1 INTRODUCTION

The widespread adoption of digital media has resulted in a significant uptick in propagating misinformation, primarily facilitated by manipulated images and captions. Cheapfakes, as a distinctive method employed for spreading deceptive information, stands out for its simplicity when compared with more complicated techniques. Specifically, Cheapfakes are recognized for their relative simplicity in information manipulation compared to Deepfakes in multimedia content manipulation.

Producing Cheapfakes entails using editing software or changing the meaning of an image or video to deceive through its captions. This manipulation of context, referred to as out-of-context (OOC) misuse of information [2], presents a unique challenge for detection, as the images and videos remain unaltered. This study's primary focus is identifying OOC image-caption pairs, with a specific emphasis on detecting the misuse of authentic photographs accompanied by conflicting image captions in news items. Due to the dangers posed by Cheapfakes and their prevalence, numerous efforts have been dedicated to Cheapfakes detection. For example, there are multiple challenges [3] [4] [6] organized to encourage researchers to contribute to this field. Built on the success of previous challenges, the Cheapfakes Detection Grand Challenge in ICMR2024 [7] has also attracted numerous researchers, fostering innovations in Cheapfakes detection.

As far as our knowledge extends, limited attention is paid to detecting cheapfakes using end-to-end networks due to the scarcity of labeled data. This paper aims to fill this gap by focusing on a comprehensive end-to-end network that operates solely on raw data, producing predictions without relying on or extracting intermediate metadata. Concretely, we strongly connect Cheapfakes detection and the Visual Entailment (VE) task. We propose a method that

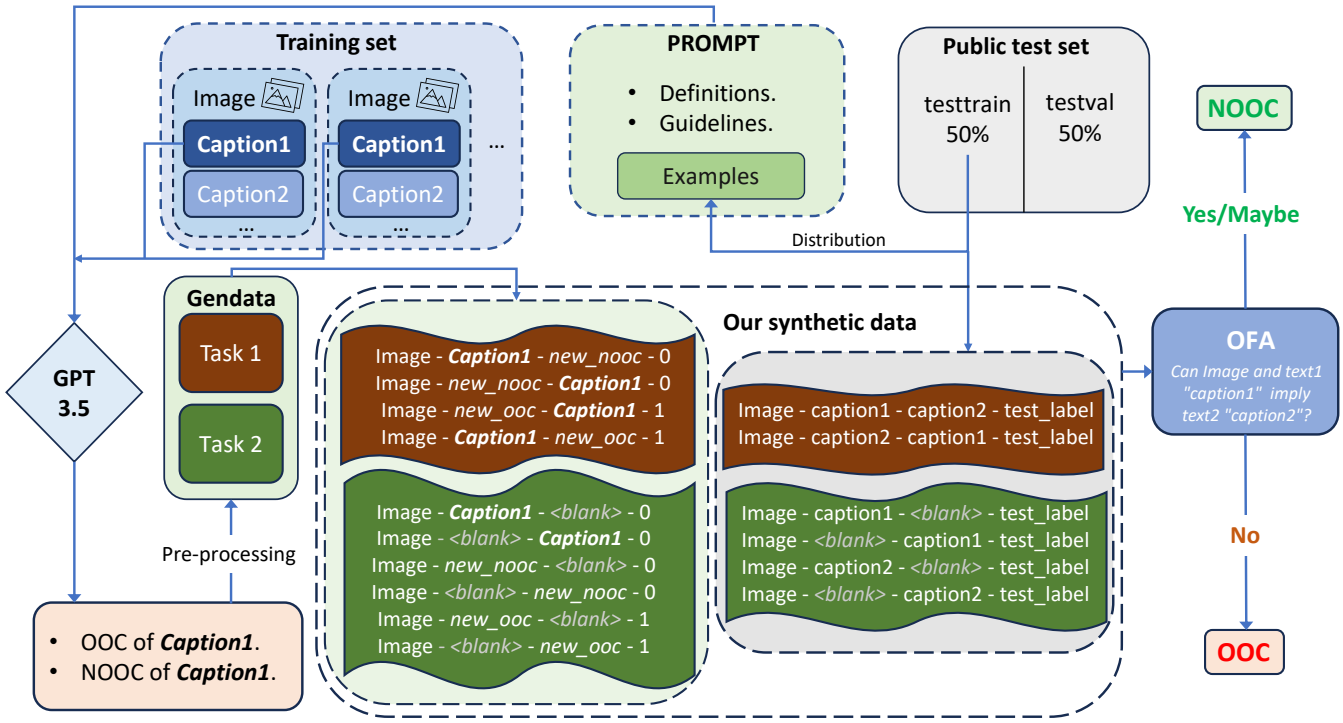


Figure 1: Pipeline of data processing and training the OFA model as our proposed End-to-end network.
We design a prompt with definitions, guidelines, and ten examples taken from the *testtrain* set. Next, we call the GPT-3.5 API from OpenAI along with *caption1* of random samples in the training set. The output includes both OOC and NOOC forms of *caption1*. Then, we format this generated dataset to match the input of 2 tasks (also for *testtrain*). Finally, all of them are fed into the OFA model. *testtrain* and *testval* are subsets from public test (see 4.1.2).

leverages VE to construct a neural network for Cheapfakes detection through Image-Text Matching and Visual Question-Answering approaches.

Given the rise of generative AI, particularly exemplified by models like ChatGPT [5], we exploit this capability to generate synthetic data for the training process. Our results indicate our proposed solution exhibits promising potential for Cheapfakes detection using end-to-end network methodologies.

Section 2 provides a concise overview of recent methods related to Cheapfakes detection. The details of our proposed method are expounded upon in Section 3, elucidating how we generate data and train our end-to-end network. The performance of our approach is evaluated using half of the public dataset of the challenge in Section 4. Finally, Section 5 concludes this work and poses questions for the future.

To summarize, in this paper, we find that:

- An end-to-end network can be useful for detecting the existence of Cheapfakes.
- Generative synthetic data from the LLM can be used to enhance the model's performance for the Cheapfakes detection problem.

2 RELATED WORK

2.1 Visual Entailment

Visual Entailment (VE) is a task in which each instance comprises a pair of an image and a sentence, whereby a premise is defined by the image rather than a natural language sentence as in traditional Textual Entailment tasks. Tran *et al.* [17] is the first work to use VE for Cheapfakes detection. However, they only used pre-trained VE as a part of their pipeline, which uses multiple parts, such as the COSMOS baseline, Natural Language Inference (NLI), etc. Wang *et al.* [19] implemented OFA, a unified pre-trained model that is useful in a variety of tasks (image captioning, visual grounding, text-to-image generation, etc.). This model also showed an efficient performance for the VE (also called image-text matching) task.

2.2 Cheapfakes detection

The awareness of Cheapfakes detection has been steadily increasing within the community. COSMOS produced by Aneja *et al.* [2] has significantly contributed by proposing the initial baseline for detecting Cheapfakes, specifically targeting the inappropriate use of authentic photos accompanied by contradictory captions in news articles.

Building upon the COSMOS dataset, Akgul *et al.* [1] have introduced four distinct strategies to enhance the precision of Cheapfakes detection. Meanwhile, La *et al.* [12] [10] [11] have suggested a novel method that involves the amalgamation of multi-modal approaches and training on select testing data; their approach involves modifying the confidence score threshold for the BERT model, enriching semantic information, and incorporating boosting techniques.

Tran *et al.* [17] have made strides in Cheapfakes detection by employing a multifaceted approach, encompassing visual entailment, online caption checking, fabricated claim detection, and natural language inference. Their comprehensive methodology demonstrates the versatility needed to spot Cheapfakes in various contexts effectively.

Furthermore, Nguyen *et al.* [16] [15] employ a multi-modal approach that seamlessly integrates object detection and natural language inference; their work becomes the pioneering effort to incorporate Named Entity Recognition in detecting Cheapfakes. Their nuanced strategy underscores a comprehensive understanding of the challenges posed by Cheapfakes and underscores the significance of adopting a multidimensional detection approach. Moreover, Dao *et al.* [8] released the first work that uses a Knowledge Graph for Cheapfakes detection. Wu *et al.* [20] is the first prompt engineering approach for Cheapfakes detection.

3 METHODS

3.1 Overview of ACM ICMR 2024 Grand Challenge on Detecting Cheapfakes

Cheapfakes, as mentioned in the section 1, is the term to refer to non-AI manipulation of multimedia. The ACM ICMR 2024 Grand Challenge on Detecting Cheapfakes is introduced to develop and figure out models that can be used to detect Cheapfakes, which means detecting whether the given image and accompanying captions are Cheapfakes (out-of-context) or not on the COSMOS dataset.

In Task 1, we will receive the triplet (image, caption 1, caption 2) as input, and our model will answer corresponding labels (out-of-context (OOC) or not out-of-context (NOOC)). That means the goal of this task is predicting the correlation of the image and the caption pair, or in other words, the existence of contextualization [7].

In the second task of the challenge, we are required to determine if a given (image, caption) pair is real or fake, the task that the average human accuracy was around 65% [14].

3.2 Adaptability of Visual Entailment

To achieve an end-to-end network for the Cheapfakes Dection, this research leverages the Visual Entailment task, which has some similar characteristics to the task of detecting Cheapfakes. On the Visual Entailment task, with OFA model [19], an image accomplished with a text premise and a text hypothesis is fed into the model's encoder. Then, the decoder will produce a corresponding label, and the model evaluates how the given image and text are semantically correlated, i.e., entailment, neutral, or contradiction. In other words, the model learns to answer "Yes", "No", or "Maybe" to the question "Can image and text 1 imply text 2?"

About the architecture, the OFA chooses Transformer [18] as the backbone architecture, with the adopted encoder-decoder as the unified architecture for all the tasks.

When performing the training process for the OFA model in the Visual Entailment task, Wang *et al.* [19] treat an image-caption pair as a positive example; then a negative example is constructed by randomly choosing another caption for the image. This is similar to the COSMOS baseline training process when Aneja *et al.* [2] consider a corresponding caption with the image and choose a random caption from the dataset to train the detection of visual correspondence. Therefore, the OFA model can be adapted for the Cheapfakes detection task.

When adapting this model for the Cheapfakes detection task, this study changes the contradiction label to out-of-context, and the entailment/neural label will be considered not out-of-context. In other words, the out-of-context label will be decided if the model's output is "no"; otherwise, it is not out-of-context.

This paper denotes the OFA model that we would train is *model*, with the input is the *image, caption1, caption2* triplet, and the output would be "yes", "no", or "maybe" label. The handle of the input triple of image and associated captions will be introduced in the section 3.3.

Our full pipeline process is depicted in figure 1, with the data processing described in the next part and the input/output of the OFA model.

3.3 Data processing

3.3.1 Data generation overview. To achieve an end-to-end solution for both Task 1 and Task 2 of the Cheapfakes detection problem, the primary obstacle is the scarcity of data. To tackle this challenge, in the era of generative AI, particularly Large Language Models (LLMs), the GPT-3.5-turbo model developed by OpenAI is chosen to leverage for generating a synthetic dataset for the OFA model. Some examples extracted from the *testtrain* set are inputted into LLMs to answer questions about their commonalities and differences. By the end of this process, we combine all of them, come up with a single prompt, and use it to generate new data.

3.3.2 Detail implementation.

- Firstly, a rule for prompt generation is established by formulating reverse inquiries to LLMs, such as "What distinguishes these two captions?" or "How can I transform caption A into caption B?". This approach enables us to extract keywords crucial for devising a generalized prompt that contains *definitions*, *instructions*, and some *examples* of the conversion.
- Secondly, the following prompt, along with *caption1* extracted from random training samples, is fed into the GPT-3.5 model to generate the necessary data for training the OFA model.

"**Definitions:**

Two captions are considered to be out-of-context (OOC) if they have most of the following differences:

1. They convey different meanings, with one refuting the claim made by the other.
2. Do not mention the same objects and subjects by name; refer to unrelated objects and subjects.

3. One caption misrepresents facts related to police violence and racial injustice, while the other does not.
4. One caption presents a direct quote attributed to a specific individual or a celebrity, while the other does not or interprets the quote in a different meaning.
5. One caption is sensationalist and entirely unrelated to the other caption.
6. If both captions mention locations, those locations should be unrelated (e.g., two locations in different countries).
7. One caption provides a straightforward description of the image or photograph, while the other is misleading, misrepresenting the facts.

Two captions are considered to be not out-of-context (NOOC) if they have the following commonalities:

1. Both captions convey the same meaning, with one being a paraphrased version of the other.
2. Both captions mention the same objects and subjects.
3. Both captions describe the same event or depict the related activities.
4. If there is a quote in one caption, the other caption should either have the same quote or a paraphrased version conveying the same meaning.
5. If both captions mention locations, those locations should either be the same or related (such as a specific location within a larger area, for example, Berlin vs Germany).
6. If both captions mention timing, those timings should either be the same or related, such as a weekday within a month, a day within a month or a year, etc.
7. If both captions mention the reason for an activity, these reasons must not conflict.

Instructions:

Given an original caption, there are two tasks you need to complete.

Task 1 (Generate OOC): you are required to generate an out-of-context (OOC) version of it with the above differences by randomly choosing 2 out of these nine steps:

1. Change the location and date to a different one.
2. Retain the concept but change its origin and content.
3. Describe different aspects of the same event with a different attitude from the original caption.
4. Randomly add a specific object reference (for example, a company name).
5. Describe and convey the opposite information about an event, a celebrity, a place, etc.
6. Remove the specific individual's name and replace it with a more generic, unrelated reference.
7. Rewrite the caption to focus on the misinformation aspect of the original caption.
8. Provide a more detailed description of the photograph's content unrelated to the original caption.
9. Negate the original caption.

Task 2 (Generate NOOC): you are required to generate a not out-of-context (NOOC) version of it with the above commonalities by randomly choosing 2 out of these six steps:

1. Add more specific details about the location and date.
2. Describe different aspects of the same event with the same attitude as the original caption.
3. Remove the specific object reference (for example, a company name).
4. Describe and convey similar information about an event, a celebrity, a place, etc.
5. Add the alternative name of the mentioned place, celebrity, event, etc.
6. Rephrase or paraphrase the original caption. If there is a quote in the original caption, the paraphrased version should convey the same meaning."

It is ensured that GPT-3.5 does not encounter any samples from the *testval* set to ensure fairness during the validation phase. To achieve this, a few examples are randomly selected from the *testtrain* set to feed into the GPT-3.5 model along with the prompt above.

Our heuristic assumes that *caption1* in the provided training set's sample is the NOOC caption with respect to its own image. This caption will accompany the above prompt and yield the OOC and NOOC captions.

3.3.3 Data structure. To train an end-to-end model for both Task 1 and Task 2, we decide to reformat the dataset in a more flexible and versatile manner as shown in the Figure 1 (*Our synthetic data* block), where label 1 and 0 mean OOC and NOOC respectively. Concretely, we acquire training data with formats such as *(Image, caption1, caption2)*, *(Image, caption2, caption1)* for Task 1, and *(Image, caption1, "")*, *(Image, "", caption1)*, *(Image, caption2, "")*, *(Image, "", caption2)* for Task 2.

Finally, we have two ways to format the generated data to accommodate both tasks. We can conduct experiments by training with generated data for Task 1, Task 2, or both simultaneously.

4 EXPERIMENTS AND RESULTS

4.1 Training settings

4.1.1 Learning rate and Number of epochs. We choose the OFA_{base} configs to proceed with the training and inference. After some experiments, we decided to keep the learning rate as the default setting provided by the OFA model's authors, $5 \cdot 10^{-5}$. In addition, we train the OFA_{base} model in 20 epochs for all datasets used for training and validation.

4.1.2 Data preparation. Along with about nearly 4000 samples of generative data, we divide the provided public test set into two subsets:

- *testtrain*: This subset contains 50% of the original public test set (500 samples) we use for training, with the class labels equally distributed between OOC and NOOC.
- *testval*: This subset comprises the remaining 50% of the original public test set we use for validation. The class labels in testval are equally distributed.

For the sake of following our experiments, here is our dataset naming convention. All datasets that combine two tasks or both *gendata* and *testtrain* are randomly shuffled.

- *gendatax*: Our generated dataset formatted for Task x.
- *testtrainx*: *testtrain* set formatted for Task x.

- *testval12*: *testval* set formatted for both Task 1 and Task 2, used to determine the efficacy of checkpoints.

To bridge the gap between the generative dataset and the provided dataset in the training network, it is noted that when the *testtrain* set is employed in the training data, the *testval* set will serve as the validation set. Additionally, we have nearly 43000 samples (nearly 40000 and 3000 samples from the generated set and *testtrain* set respectively) for training by using formatting strategies defined in Figure 1.

4.2 Inference process

We use the provided public test set to evaluate the model after training for the inference process. There are three labels returned from the network, if the returned label is 0, the image-caption pair will be classified as out-of-context, otherwise, the pair will be classified as not out-of-context. Instead of traditional methods of inference only in one way ($model\,(image, caption1, caption2)$ or $model\,(image, caption2, caption1)$), we propose a bi-directional inference process, as an ensemble method, what is explained below.

First, we will get the answer from the network using the following formula:

$$\begin{aligned} output_1 &= model\,(image, caption1, caption2) \\ output_2 &= model\,(image, caption2, caption1) \\ answer &= \text{argmax}\,(output_1 + output_2) \end{aligned} \tag{1}$$

Each of $output_1$ and $output_2$ is a vector of 3 elements (c, e, n), with c, e, and n represent for contradiction, entailment, neutral, respectively. As aforementioned, if the label is *contradiction*, the data point will be considered out-of-context. Otherwise, it will be treated as not out-of-context. Therefore, the following formula is proposed to obtain the *label* from *answer*.

$$label = \begin{cases} \text{OOC if } answer \text{ is } 0 \\ \text{NOOC, otherwise} \end{cases}. \tag{2}$$

The inference result will be given in Table 1.

4.3 Inference results

4.3.1 Effectiveness results. Table 1 displays the experiment results for Cheapfakes detection Task 1 and 2. The first row of the table is our results for Task 1 and Task 2 on the public test, private test 2023, and private test 2024. As inferred from the table, our method with a unified network performs very effectively on the public test, which 95.60% for Task 1, and 93.00% for Task 2, compared with 82.00% of the COSMOS baseline, which achieves a hold of only 82.00% on Task 1 [2]. On the private test of 2023, the network shows a high accuracy of 88.64% for task 1, which is still far higher than the COSMOS, and 65.50% for Task 2, which is better than human-level performance [14]. The remaining rows in Table 1 show the results of other participants, to compare the performance with our end-to-end network.

Moreover, the second column in Table 2 shows different accuracy of different OFA models, trained with *gendata1*, *gendata2*, and the chosen model, trained with *gendata1*, *testtrain1*, and *testtrain2*, to have a comparison and make decision of the chosen model. This column contains two values separated by a slash, which are accuracy on Task 1 and Task 2, respectively. From the table, it can be concluded that, with only the *gendata1* and *gendata2* for training, the OFA model performs an accuracy of 84.50% on Task 1, which is a better number than the COSMOS baseline.

Additionally, to bridge the gap between the generated data and the public test, to make the model more general with the data, half of the public test (*testtrain1*, *testtrain2*) is added to the training set, and the model achieves an accuracy of 95.60% for Task 1 and 93.00% for Task 2. For a fair comparison, the chosen model is also evaluated on the *testval* set, which is not included in the training process, and gives a notable result of 91.20% and 86.20%, which is still much better than the COSMOS baseline.

4.3.2 Efficiency results. The efficiency results are shown in Table 2, which includes three efficiency metrics of our proposed method. Our average GFLOPs are 280 for Task 1 and 257 for Task 2, the model size is 2.44 gigabytes, the number of parameters is 182 million, and latency is 174ms for Task 1 and 161 ms for Task 2. Regarding the latency metric, for a fair comparison, we run both our proposed method and COSMOS baseline on the same environment, equipped with a core i7 10700 CPU, an RTX 3060 GPU with 12GB VRAM, and 48GB DDR4 RAM, see Table 2. Although the COSMOS baseline's latency is lower than our proposed method, it requires running at least two additional models, such as object detection and text encoder, in case detecting misinformation in new samples is required.

5 CONCLUSION AND FUTURE WORK

In summary, the term Cheapfakes detection will become increasingly important in the future, given the advancement of digital society, where the context of an image can be easily manipulated for malicious purposes. Our paper proposes a method for Cheapfakes detection using OFA and generative AI for data augmentation, particularly an end-to-end and unified network, with a notable result, for both Task 1 and Task 2.

The competitive results of our approach claim that it has an essential role in practical situations for detecting Cheapfakes. For example, it can be used for social media platforms to develop a system to delete potential image miscontextualization automatically.

We look forward to the continued improvements of this approach and the more efficient solutions for the Cheapfakes detection task. As technology is growing with unbelievable velocity, it is vital to continue to research and develop innovations for Cheapfakes detection.

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Table 1: Inference results

Method	Task 1 - Public	Task 2 - Public	Task 1 - Private 2023	Task 2 - Private 2023	Task 1 - Private 2024	Task 2 - Private 2024
Ours	**95.60%**	**93.00%**	**88.64%**	**65.50%**	61.69%	45.16%
Vo-Hoang et. al.	86.00%	—	80.90%	—	62.50%	—
Pham et. al.	88.90%	—	87.70%	—	**72.20%**	—
Le et. al.	79.40%	—	71.36%	63.00%	52.82%	**54.84%**
Vu et. al.	82.90%	—	72.27%	—	64.52%	—
Seo et. al.	71.90%	55.70%	75.90%	61.50%	64.11%	50.81%

The inference result of the ACM ICMR 2024 Grand Challenge on Detecting Cheapfakes. The first column is the author of the method, and the second and the third column is the accuracy on the public test set of task 1 and task 2, respectively. The fourth and fifth column is the accuracy for task 1 and task 2 for the private test of 2023, while the sixth and seventh column is the accuracy for task 1 and task 2 of the private test of 2024.

Table 2: Efficiency results and comparison between trained models

Method	Accuracy (Task1/Task2)	Accuracy on testval	GFLOPs (Task1/Task2)	Model Size	# Params	Latency (Task1/Task2)
OFA (not trained on gendata)	52.00%/49.50%	—	280/257	2.44GB	182M	
OFA (trained on gendata1)	72.40%/53.10%	—				
OFA (trained on gendata2)	78.90%/57.50%	—				
Our submission model	95.60%/93.00%	91.20%/86.20%				174ms/161ms
COSMOS				10.2MB	2.5M (trainable)	136ms/ —

Table of efficiency results and comparison between different trained OFA models to find the best one. The submission model is trained on *gendata1*, *testtrain1*, and *testtrain2*, and its accuracy is evaluated on both the whole public test and the *testval*.

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